Task 3: Modelling Budgets and Downside Risk

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1 Task 3: Modelling Budgets and Downside Risk

1.1 Introduction

Meridian's Finance team needs two inputs for planning: a realistic probability model for production budgets, benchmarked against Universal Pictures, for its budget simulator; and a downside-risk figure for Action, its most capital-intensive genre. Downside risk is the 25th percentile of ROI (worldwide box-office revenue divided by production budget), the multiple below which the worst quarter of Action films fall.

1.2 Universal Pictures Budget Modelling

[Document the distributions considered.]

1.3 Model Diagnostics and Comparison

Distribution	Parameters (units)	Estimate 1	Estimate 2	Log-likelihood	AIC
Gamma	shape, rate (per $m)	1.382	0.031	-1,158.1	2,320.1
Weibull	shape, scale ($m)	1.214	47.479	-1,158.3	2,320.6
Log-normal	meanlog, sdlog (log $m)	3.392	1.015	-1,172.6	2,349.2
Normal	mean, sd ($m)	44.481	37.432	-1,225.1	2,454.1

Table 1: Distributions fitted to Universal Pictures budgets, ordered by AIC (lowest first)

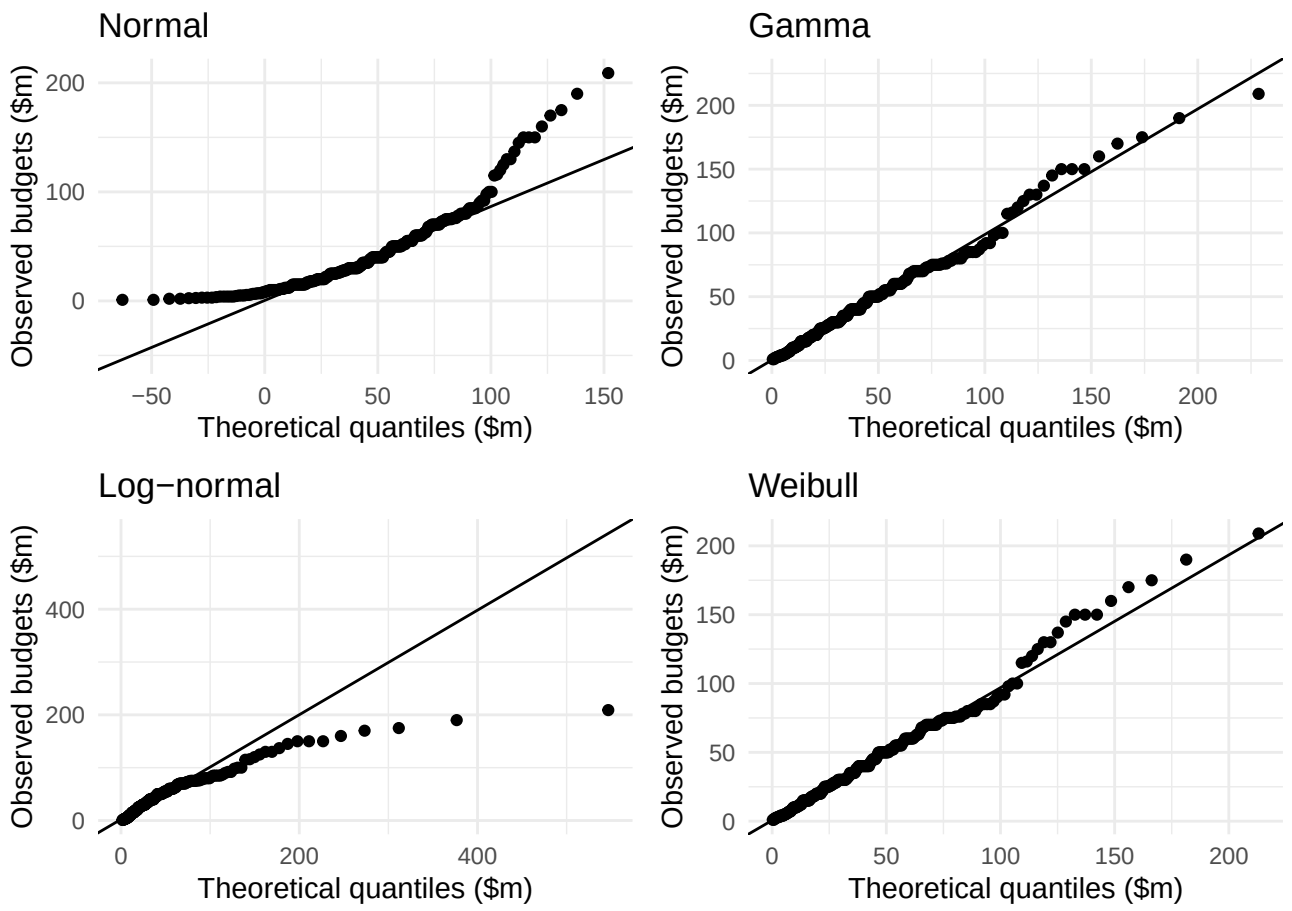


Figure 1: Q-Q plots of Universal Pictures budgets against each fitted distribution. Points close to the line indicate a good fit.

1.4 Action Genre Downside Risk

Method	25th percentile of ROI	95% lower	95% upper
Percentile bootstrap	0.91	0.80	1.05
Fitted log-normal model	0.73	0.66	0.82

Table 2: 95% interval estimates for the 25th percentile of ROI for Action films, from two methods

1.5 Comparison of Interval Estimates

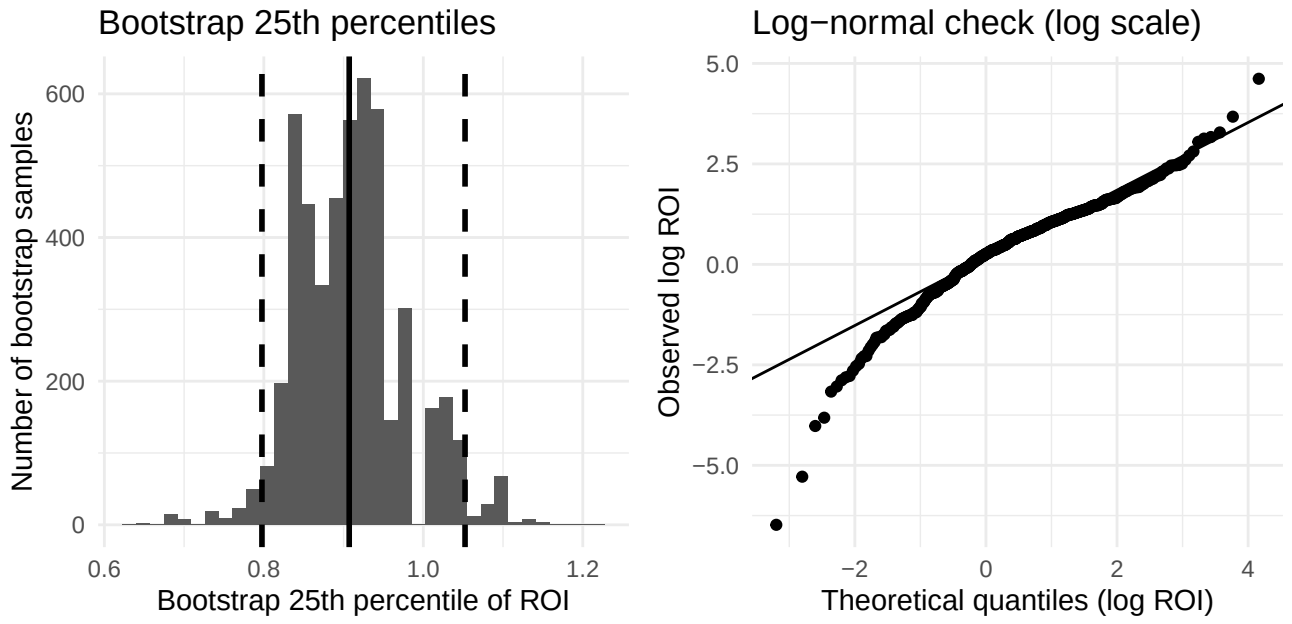


Figure 2: Left: bootstrap distribution of the 25th percentile of Action-film ROI (solid line: sample estimate; dashed lines: 95% interval). Right: Q-Q plot of log ROI against the fitted log-normal model.

1.6 Summary for the Finance Team